Running log — improvements to the class-AB pad driver

`sg13cmos5l_cm_ip__single2diff2single`, `design_considerations/class_ab_pad_driver/improvements`

2 October 2026

Sessions 2026-10-02, 2026-10-03 and 2026-10-05, Claude (configured model `claude-opus-5-5`; the serving model may differ). Newest entries at the bottom. Times are Europe/Berlin, taken from the session transcript (corrected 15:01: the first versions of several headings carried guessed times, up to $1\frac{1}{4}$ h too late).

Task as given: area, current and loop gain of `d2s_miller` / `d2s_loadcomp` are unsatisfactory, probably because of the linear passives (rhigh degeneration, MOM Miller caps). Look in Edwards, Minch, Cauwenberghs, Hasler (DoNotLitter/TimEdwards, LogDomain, FloatingGate) and Sarpeshkar 1997 for (1) alternatives to resistor source degeneration in the DDA, (2) compensation without linear BEOL capacitance, (3) cascoding with the output devices either doubling as ESD clamps or as separate devices in the slot, (4) externally-linear / internally-nonlinear circuits whose internal nonlinearity is not the subthreshold exponential. End result: proposed improvements to `class_ab_pad_driver`.

Nothing is written to `~/DoNotLitter`. Papers are cited, not copied (chatlog/ref rule).

14:10 — baseline taken from the existing note and the first CACE run

What the CACE run (`verification/cace/first_run_summary.md`) adds to the design note:

- Gain spread 0.476…0.554 over PVT + rhigh corners, 0.41…0.61 in process MC. Cause: the gain is (R2 + 2/gm)/(R1 + 2/gm); R tracks R, but gm does not track R.
- Output offset in mismatch MC −32…+49 mV (4 degenerated pairs; offset referred through 1/Gm of degenerated pairs is large).
- INL 469 mV worst corner; THD −15 dB at 1.25 V output (pairs run out of I_tail·R range).
- Output noise 1.6 µV/$\sqrt{\text{Hz}}$ at 1 kHz (degeneration resistors + low Gm).
- Area: the two MOM Miller caps (2 × 31 × 31 µm) dominate the non-frame area (1920 of ∼2500 µm²).
- Current: 427 µA, of which 240 µA output-stage quiescent, ∼180 µA front end + bias.

14:25 — literature read so far (details in `literature_notes.md`)

- Sarpeshkar 1997 thesis (scanned, OCR'd locally on the linked computer), ch. 3 = the wide-linear-range OTA: well input, diode source degeneration, gate degeneration through the mirror diode, bump linearization (w = 2 cancels the cubic term). No resistors.
- Edwards thesis §3.5–3.10, Edwards & Cauwenberghs 2000: ELIN/log-domain; names the square law of strong inversion as the other usable internal nonlinearity; matching = "same surround".
- Minch thesis ch. 7: a source follower is linear whatever the device law (MITE via FG source follower, valid above threshold).
- Hasler thesis ch. 4–5, Sánchez-Sinencio FG slides: capacitive dividers / FG inputs widen linear range, but need linear caps and charge control (tunneling/injection or UV).

14:25 — simulation environment

Cloud container: ngspice-42 (Ubuntu), IHP-Open-PDK `dev` sparse checkout, OpenVAF-reloaded built from source with LLVM 18 for psp103 / r3_cmc / cap_cmomi. (in progress)

14:28 — simulation environment ready, baseline reproduced

ngspice-42 + IHP-Open-PDK `dev` models + psp103 / r3_cmc / cap_cmomi compiled with OpenVAF-reloaded (LLVM 18, OSDI minor version patched 4 → 3 for ngspice 42, as in the first session). `run_d2s.py` functions reproduce `results.txt` exactly (A, 1 kΩ ∥ 100 pF: gain 0.5005, 0.18 mV, nonlin 5.52 mV, T0 66.5 dB, fc 1.62 MHz, PM 74.8°, THD 10 kHz −51.1 dB).

14:40 — DDA unit characterization (`sim/units.spice`; test bench now in `run_improvements.py units`)

Each unit is one split-tail PMOS pair; x = gp − gn swept ±0.8 V at input CM 1.40 / 1.65 / 1.90 V.

unit	degeneration	Gm0 @ CM 1.40 / 1.65 / 1.90	f(0.5)/(Gm0·0.5)	CM sensitivity of Gm
unit_r	rhigh 0.5 × 36.4 µm (108 kΩ), I_t = 2 × 10 µA	15.54 / 15.55 / 15.58 µS	0.996	0.5 %/V
unit_t	hv PMOS 0.5/3 in triode, gate at vss	12.43 / 15.54 / 18.40 µS	0.992	77 %/V
unit_w	Sarpeshkar well input, 4/4, gates at vss, 1 × 20 µA	11.82 / 11.02 / 10.35 µS	0.991	−27 %/V

The MOS alternatives are linear enough on their own; their problem is common-mode sensitivity (triode R depends on V_SG = source − gate; well-input κ depends on well-to-gate voltage, as Sarpeshkar's §3.3.1 reports). Why that matters for the next step: see 14:43 (a leftover "15:05" from the guessed timestamps, corrected 16:15).

14:43 — matched-pair DDA (MP-DDA)

Idea: four identical units, A (vinp, vref), B (vref, vinn), C1 = C2 (vref, vfb): f(vinp − vref) + f(vref − vinn) = 2 f(vfb − vref). With matched units the solution is vout − vref = (vinp − vinn)/2 for *any* odd monotonic f (exact for input CM = vref; for a CM error c the error is second order in c). Gain $\frac{1}{2}$ = unit count ratio; no R1/R2, no gm term. This is static ELIN: the input units compress, the feedback units invert the same law. Caveat found analytically: A and C sit at CM vref + x/2, B at vref − x/2. A unit whose Gm depends on CM (k per volt) gives v ≈ u − k u²/2 → unit_t ≈ 96 mV, unit_w ≈ 34 mV of second-order error at u = 0.5 V, unit_r ≈ 0.6 mV. So in this topology the poly resistor stays the right degenerator. Its area is ∼18 µm² per unit, it never was the area problem.

First simulation, everything else unchanged (fold, mirror, class-AB, frames, MOM caps, bias):

variant	load	gain	nonlin	T0 / fc / PM	THD 10 k / 100 kHz
baseline A	1 k ∥ 100 p	0.5005	5.52 mV	66.5 dB / 1.62 MHz / 74.8°	−51.1 / −50.2 dB
MP-DDA, MOM	1 k ∥ 100 p	0.4998	1.04 mV	66.5 dB / 1.62 MHz / 74.7°	−60.2 / −55.5 dB
MP-DDA, MOM	100 p	0.5000	1.02 mV	96.5 dB / 2.03 MHz / 66.2°	−60.0 / −59.8 dB

variant	load	gain	nonlin	T0 / fc / PM	THD 10 k / 100 kHz
MP-DDA, hv-PMOS accumulation cap 14 × 14 µm	100 p	0.5000	1.02 mV	96.7 dB / 2.80 MHz / 53.4°	−60.0 / −59.9 dB

Same current (427 µA) and tail budget (8 × 10 µA = baseline 4 × 20 µA). 9 dB better THD, 5× lower DC nonlinearity. The MOS cap (196 µm² vs 961 µm² MOM) does not degrade THD despite a 2× C–V variation over 0.15…1.25 V (0.37 → 0.75 pF); it is just smaller than 1 pF on average, so it needs resizing for the PM.

14:48 — MOS capacitor sizing, corners, mismatch

MOS accumulation cap size (MP-DDA, f_c / PM): 18 × 18 µm matches the 31 × 31 µm MOM pair almost exactly (100 pF: 2.00 MHz / 67.0° vs 2.03 MHz / 66.2°; 300 pF: 46.3° vs 45.5°; 1 nF: 26.9° vs 26.4°). Area per side 324 µm² instead of 961 µm² (3.0×), no metal used.

Corners (dc gain fitted over $|vd| \leq 0.2$ V, 1 kΩ ∥ 100 pF; nonlin = max deviation over ±1 V):

corner	baseline A gain / nonlin	MP-DDA gain / nonlin
tt 27	0.5005 / 5.52 mV	0.4998 / 1.04 mV
ss 125 (res wcs)	0.5119 / 8.07	0.4988 / 1.56
ff −40 (res bcs)	0.4968 / 3.30	0.4998 / 0.66
sf / fs 27	0.4998 / 4.11, 0.5013 / 8.07	0.4998 / 0.82, 0.4997 / 1.37
ss 27 / ff 27	0.5022 / 12.69, 0.4990 / 3.30	0.4998 / 1.91, 0.4997 / 0.69
tt res wcs / bcs	0.4895 / 7.09, 0.5148 / 5.07	0.4997 / 1.48, 0.4998 / 0.66
tt 125 / tt −40	0.5249 / 5.18, 0.4870 / 7.15	0.4983 / 1.10, 0.4998 / 1.53

Systematic gain spread −2.6…+5.0 % → −0.34…0 %. Loop gain / PM unchanged. (Corrected 15:25: this line first said "±3.8 %", which was half the spread, not its range.)

Mismatch MC (30 seeds, `mos_tt_mismatch` + `res_typ_mismatch`, mm_ok=1 on all MOS and rhigh): baseline offset σ 14.4 mV, MP-DDA σ 14.5 mV — the front-end change does not touch the offset. Budget by device group (mismatch on one group at a time, 20 seeds): folding sinks 11.5 mV, tails 4.2, mirror 2.3, pairs 1.7, everything else < 0.4. Same mechanism Sarpeshkar describes (§3.3.4: current mismatch / transconductance; low Gm/I front ends are offset-prone, and the mirrors, not the input devices, matter most).

14:53 — `d2s_mp2`: same topology, resized for current and offset

- units at 2 × 5 µA (long-L tails 5/6), R = 150 kΩ (rhigh 0.5 × 50.6 µm), I_t·R = 0.75 V
- folding sinks 30 µA, 36/6 (lower gm/I), mirror 24/4 at 10 µA
- bias fixture `d2s_bias_lp`: reference currents 5 + 5 + 2 + 2 + 5 + 5 µA instead of 20 + 50 + 5 + 5 + 5 + 5
- headroom limit found on the way: a ≈ 2.6 V and b ≈ 0.7 V are the output-device gates, so the mirror and the sinks each get only ∼0.35 V; their overdrive can't be raised for matching, only their area.
- all devices ≥ 70 mV from saturation edge at vd = 0 (`oppoint.py`)

	baseline A	d2s_mp2, MOS 16 × 16
I_DD (incl. bias fixture)	427 µA	297 µA
I_Q P/N	240 µA	213 µA
compensation area	2 × 961 µm² MOM (M1–M4)	2 × 256 µm² hv-PMOS
T0 1 k ∥ 100 p / 100 p / 50 Ω	66.5 / 96.5 / 40.7 dB	69.7 / 99.6 / 43.9 dB
fc / PM, 100 p	2.02 MHz / 66°	1.76 MHz / 63°

	baseline A	d2s_mp2, MOS 16 × 16
THD 10 kHz (1 k ‖ 100 p)	−51.1 dB	−58.7 dB
THD 100 kHz (1 k ‖ 100 p)	−50.2 dB	−51.4 dB
nonlin (±1 V vd)	5.52 mV	1.24 mV
offset σ (mismatch, 30)	14.4 mV	5.3 mV
gain σ (mismatch, 30)	1.6 %	1.1 % (0.84 % from rhigh unit mismatch)

Gain σ is now set by rhigh area (0.5 × 50.6 µm per unit): 4× the area halves it.

14:57 — the right figure of merit for a DDA unit (`sim/run_improvements.py units`)

The first unit test swept both inputs around a common CM. In the MP-DDA one input of every unit sits at vref, so the relevant comparison is fA(x) = I(vref + x, vref) against fB(x) = I(vref, vref − x): the same differential voltage, but the pair's sources sit x lower in B. Solving fA(u) + fB(u) = 2 fA(v) from the unit curves alone predicts the whole driver's static error; for unit_r it gives −1.04 / −1.02 mV at u = ∓0.5 V, which is the full-circuit 1.04 mV.

unit	own compression at 0.5 V	MP-DDA error at u = −0.5 / +0.5 V
unit_r (split tails + rhigh)	0.5 %	−1.0 / −1.0 mV
unit_t (triode PMOS)	20 %	−191 / −72 mV
unit_w (well input)	7 %	+29 / +38 mV
unit_q (undegenerated 2/6, square law, no R)	22 %	+75 / +200 mV

So the criterion for an ELIN unit here is not its linearity but that its output depends on gp − gn only. The split-tail pair meets it because each device is a constant-current source follower (Minch thesis ch. 7: a follower is linear whatever the device law) and the element between the sources sees exactly gp − gn. Any element whose value depends on the absolute source voltage (triode R, well κ, tail/CLM of a single-tail pair) fails.

14:56 — cascoding

(a) output devices = ESD clamp frames. A series cascode would put its drain, not the clamp's, on the pad; a stacked-clamp frame would be a new ESD structure to qualify. So no output cascode. What remains is cascoding the internal high-impedance nodes. Lengthening the folded and mirror cascodes (L 1 → 3 µm, W scaled, same overdrive) in `d2s_mp2`:

cascode L	T0 1 k ‖ 100 p	T0 100 p	T0 50 Ω	PM 100 p
1 µm (d2s_mp2)	69.7 dB	99.6 dB	43.9 dB	63.4°
2 µm	75.4	103.1	49.6	62.8°
3 µm (d2s_mp2c3)	78.9	104.3	53.1	62.0°

+9 dB for ∼320 µm² of gate area. Regulated (gain-boosted) cascodes were not tried: node a sits at ∼2.6 V and b at ∼0.7 V (the output gates), which leaves ∼0.35 V per device for the mirror, the sinks and their cascodes, so no headroom for an auxiliary amplifier's V_GS.

(b) separate output devices in the slot. Then they can be cascoded. Test case `d2s_lc2`: load-compensated (topology B, no compensation capacitor at all), MP-DDA front end, half-frame output devices, cascode gates simply at vref (output range ∼1.0 . . . 2.4 V):

	T0	5 p	20 p	100 p	1 n	1 k ∥ 100 p	I_DD
no cascode	37.2 dB	16°	47°	78°	89°	no loop gain	174 µA
cascoded	74.0 dB	unstable	39°	77°	89°	no loop gain	169 µA

Cascoding adds 37 dB; a resistive load still collapses a single-stage loop, and the minimum load capacitance stays (∼30–40 pF). In the slot, the pad goes through IOPadAnalog, whose `padres` is behind 587 Ω (SecondaryProtection). Sensing at `padres` (near side) makes that resistor an isolation resistor: PM at 20 pF 39° → 62°, 100 pF → 104°; the pad then sees a 587 Ω · C_L low-pass outside the loop (2.7 MHz at 100 pF). Sensing at `pad` keeps the PM of the no-resistor case. 50 Ω through 587 Ω is not drivable at all. Open point: the cascoded version has 7.8 mV systematic offset (diode replicas at a different V_DS than the cascoded output devices); cascoding the diode replicas would fix it.

14:58 — noise (1 k ∥ 100 p, output-referred)

	1 kHz	100 kHz	10 Hz–10 MHz
baseline A	1599 nV/√Hz	200 nV/√Hz	278 µV rms
MP-DDA (baseline sizing)	1592	201	279
d2s_mp2c3	676	152	231

14:59 — `d2s_mp2c3` (= d2s_mp2 with L = 3 µm cascodes, MOS caps 16 × 16 µm): full table

Renamed `d2s_mpdda` (`sim/d2s_mpdda.spice`); every number re-run in `sim/results_improvements.txt`. 296 µA (baseline 427), T0 78.9 dB at 1 k ∥ 100 p (66.5), 53.1 dB at 50 Ω (40.7), 104 dB capacitive (96.5), PM 62° at 100 pF, 42° at 300 pF. Gain 0.4992…0.5000 over 11 corners, nonlin ≤ 3.5 mV (ss 27 °C). Offset σ 5.3 mV, gain σ 1.1 % (rhigh area). THD 10 kHz −58.5 dB; 100 kHz −51.3 dB (1 k ∥ 100 p; lower fc than the MOM-MP version, 1.37 vs 1.62 MHz).

15:05–15:20 — consolidation

Intermediate netlists and scripts folded into one runner, `sim/run_improvements.py`, which reuses `../run_d2s.py` unchanged and reproduces every number of this log and of `proposed_improvements.md` in one run (3 min 49 s, `sim/results_improvements.txt`). The re-run matches the numbers above. Files in `sim/`:

file	content
`units.spice`	DDA units: unit_r (rhigh), unit_r2 (final, 2 × 5 µA), unit_t (triode MOS), unit_w (well input), unit_q (undegenerated)
`d2s_mp.spice`	MP-DDA in the baseline's sizing; compensation through the `ccomp` wrapper
`ccomp.spice`	ccomp_mom (as d2s_miller), ccomp_mos (hv PMOS accumulation caps)
`d2s_mpdda.spice`	the proposal (frames kept as output devices)
`d2s_bias_lp.spice`	bias fixture with 26 µA of reference current instead of 90 µA
`d2s_lc2.spice`	case (b): in-slot output devices, load-compensated, cascoded (and not)
`run_improvements.py`, `results_improvements.txt`	runner and its output

15:18 — transfer check

Checksums of the committed files against the container copies: `proposed_improvements.md`, `sim/results_improvements.txt` and `sim/run_improvements.py` did not match after the first commits, although the tool reported "written". The device still held earlier versions of them, 20530 / 16367 / 13939 bytes instead of 20571 / 18250 / 16300. Re-committing from the same staged paths, with force, changed nothing. Committing copies from a *new* staged directory (`outputs/xfer2/`) worked. All files now match by sha256. This looks like the silent no-op described in the project CLAUDE.md, §6. Workaround: stage every new version of a file under a new staged path.

15:20 — time and tokens used by this task

Wall clock: 14:08 → ∼15:20, about 72 minutes, interrupted twice by questions. Of that, the compute (not the model) took about 3 min for the OpenVAF build and model compile, about 6 min for OCR on your computer, and about 10 min of ngspice in total, including the 3 min 49 s full re-run.

Tokens, from this session's own transcript (`~/.claude/projects/-home-claude/<session>.jsonl`, per API response, as of 15:19):

phase	wall clock	API calls	output tokens	cache writes	cache reads
orientation: existing note, netlists, CACE results, CLAUDE.md	14:08–14:12	14	20,138	106,368	1,958,053
simulation environment + literature reading/OCR, interleaved	14:12–14:27	49	23,126	122,848	12,499,286
baseline reproduced; Caltech-thesis check	14:27–14:31	6	2,887	13,684	1,960,595
design reasoning + simulations	14:31–15:00	40	125,771	157,464	17,134,286
accounting, log timestamp correction	15:00–15:02	5	10,264	17,604	2,485,393
consolidated runner, re-run, notes, proposal, transfer checks	15:02–15:19	37	60,208	98,924	21,839,596
total, main thread		151	242,394	517,090	56,877,712
subagent (Caltech PDF check)	14:30–14:31	5	1,463	94,882	368,538

How to read it:

- **Output tokens** (∼244 k): what the model wrote, including its reasoning. More than half of it falls in the design phase.
- **Cache reads** (∼57 M): each API call re-reads the whole conversation so far from the prompt cache. The context grew from ∼118 k tokens at the start (system prompt, tools, memory) to ∼608 k at the end. The growth came mostly from tool output: paper text, OCR, netlists and simulation listings.
- **Cache writes** (∼0.5 M): new context added to the cache.
- **Uncached input**: 302 tokens in all.

How cache reads, cache writes and output count against a specific plan's usage limits is not something this log can state (not checked here); the plans' usage limits are documented at support.claude.com. What this kind of task pushes is the context size and the number of calls made over it.

What drove the size, for planning similar tasks:

- Reading papers as text into the conversation: Edwards, Minch and Hasler excerpts and the Sarpeshkar OCR, roughly 100–150 k tokens of context.

- Long simulation listings.
- A long reasoning phase before the first simulation.

A second session that starts from `proposed_improvements.md` and `log.md` instead of the papers would begin with far less context.

15:55 — IEEEtran version

`proposed_improvements.tex` is `proposed_improvements.md` in IEEEtran journal style: the same content and numbers, with equations and a references list taken from `literature_notes.md`. `proposed_improvements.pdf` (5 pages) was built in the cloud with pdflatex, run twice, and had no errors and no overfull boxes. The only packages it needs besides IEEEtran.cls are amsmath, booktabs, tabularx, array, textcomp, url and cite; there is no siunitx. The sandbox I reach on the linked computer has pdflatex but neither IEEEtran.cls nor siunitx. Building there needs TeX Live's `texlive-publishers`, or a copy of IEEEtran.cls next to the file.

16:06 — time and tokens, updated through the LaTeX work

Same method as at 15:20, now including the report, the questions afterwards and the two LaTeX conversions:

phase	wall clock	API calls	output tokens	cache writes	cache reads
orientation	14:08–14:12	14	20,138	106,368	1,958,053
simulation environment + literature/OCR	14:12–14:27	49	23,126	122,848	12,499,286
baseline reproduced; Caltech-thesis check	14:27–14:31	6	2,887	13,684	1,960,595
design reasoning + simulations	14:31–15:00	40	125,771	157,464	17,134,286
accounting, log timestamp correction	15:00–15:02	5	10,264	17,604	2,485,393
runner, re-run, notes, proposal, transfer checks, report	15:02–15:20	41	64,333	102,341	23,278,772
questions about the spreadsheet button	15:20–15:52	1	427	6,984	613,327
IEEEtran LaTeX of the proposal	15:52–16:04	16	32,152	45,403	10,226,268
resource-log question; start of the log's LaTeX	16:04–16:06	2	1,576	7,552	1,339,858
total, main thread		174	280,674	580,248	71,495,838
subagent (Caltech PDF check)	14:30–14:31	5	1,463	94,882	368,538

The context is now ∼674 k tokens per call. The LaTeX conversion of this log comes after this table and is not in it.

16:15 — LaTeX version of this log

`log.tex` / `log.pdf` (article class, not IEEEtran, in the style of `sudelbuecher/esd_protection/*.tex`). The body was converted from this file with pandoc 3.1.3, and the tables were given widths weighted by their content. Builds with pdflatex using only standard packages (booktabs, longtable, calc, hyperref, xurl, microtype, textcomp, geometry). A snapshot: entries after 16:15 are only in `log.md`.

16:37 — testbench decks in sim/tb/

Nine stand-alone ngspice decks, so the results can be rerun without run_improvements.py: tb_mpdda_dc, tb_mpdda_loop, tb_mpdda_step, tb_mpdda_thd, tb_mpdda_noise, tb_mpdda_op, tb_units, tb_moscv and tb_lc2_loop. Each one names its reference value from results_improvements.txt in the header and runs from its own directory (ngspice -b <deck> prints, ngspice <deck> plots). They need the PDK .spiceinit (sourcepath to the models, OSDI for psp103, r3_cmc and cap_cmomi). Rerun in the cloud, they reproduce the reference: gain 0.49994, offset 0.055 mV, nonlinearity 1.243 mV, Idd 296 µA; loop 78.9 dB, 1.366 MHz, 72.4°; THD 0.119 %; noise 676 / 152 nV/$\sqrt{\text{Hz}}$ at 1 / 100 kHz; d2s_lc2 near side 74.0 dB, 1.49 MHz, 104.3°.

17:00 — xschem schematics in xschem/

First drafts for hand editing, laid out like the hand-edited ../xschem/d2s_miller.sch: ports in one column on the left, output on the right, vdd and vss as wires across the sheet, one column per current branch, bulks wired, and the input and bias nets as wire buses from the port column. Internal nets carry lab_wire labels so that the netlist keeps the source's net names.

- unit_r2 (the unit of d2s_mpdda), and unit_r, unit_t, unit_w, unit_q. All five share one box symbol: gp and gn on the left, vdd and vbp on top, x, y and vss at the bottom.
- d2s_mpdda: four unit_r2 boxes; the fold, mirror, class-AB pair and output devices sit where they are in d2s_miller.sch. CMA and CMB are drawn as hv PMOS accumulation capacitors.
- d2s_lc2 and d2s_lc2_nc: the same core. The output cascode gates reach vref through one lab_pin.
- d2s_bias_lp, and d2s_mpdda_biased, the CACE fixture, which is wired like d2s_miller_biased.sch.
- The symbols of d2s_mpdda, d2s_lc2(_nc), d2s_bias_lp and d2s_mpdda_biased are copies of d2s_miller.sym, d2s_bias.sym and d2s_miller_biased.sym, which have the same port lists.

Parameters are frozen at the .subckt defaults:

- lcas = 3, so CX, CY, PCL and PCR are 30u / 3u with ng = 2.
- wc = lc = 16u.
- rl = 50.6u in unit_r2 and 36.4u in unit_r.
- iab = 5u, ibnc = 2u, wdp = 6.66u, wdn = 3.8u.
- unit_w keeps w=wwi l=lwi, which the testbench defines.

Not drawn: d2s_mp, whose ccomp wrapper exists only in the testbench, and ccomp_mom / ccomp_mos; ccomp_mos appears in d2s_mpdda as CMA and CMB.

check_xschem.py netlists every .sch with xschem and compares each subcircuit in the netlist with ../sim/*.spice, device by device. It checks model, nets in order, w / l / ng / value, and the port order of both .sch and .sym. With xschem 3.4.4 it finds 0 mismatches over the 10 cells, including the unit_r2 bodies inside the parents. It did report each deliberate error I introduced: a changed port label, a resistor length, a current value and two swapped .sym pins.

In the first draft, nets l2 and b came out split (net1, net2). A bus crossed the middle of a column without a junction, and xschem connects a wire only at its end points. Hand edits can break connectivity the same way, so rerun check_xschem.py after editing. Most of the 106 wire crossings in d2s_mpdda are in the input and bias bus band under the four units; that band is the obvious place for hand rearrangement or for labels. The script that drew the sheets stays in the cloud, since the drafts are meant to be edited by hand from here on.

17:15 — LaTeX log brought up to date

log.tex and log.pdf were regenerated from this file through this entry, with the same steps as at 16:15 (pandoc, then the table widths). They are still snapshots: entries after this one are only in log.md until the next regeneration.

17:35 — block annotations, flat schematics, xschem/README.md

Every sheet now outlines its functional blocks with magenta dashed boxes tagged [1] to [9], with a legend under the sheet. A number means the same block on every sheet: [1] DDA units, [2] fold, [3] mirror, [4] class-AB control, [5] its copy in the mirror input branch, [6] output devices, [7] Miller capacitors (cascodes in d2s_lc2), [8] bias, [9] diode replicas in d2s_lc2. xschem/README.md explains the blocks with sizes, nets and tt currents, gives the signal path and the bias network, and lists the files.

Two new sheets show everything at transistor level:

- d2s_mpdda_flat: d2s_mpdda with the four units drawn out. Unit devices and nets carry the unit's name (Ta_A, sa_A, ...). It has the same 13 pins.
- d2s_mpdda_bias_flat: d2s_mpdda and d2s_bias_lp on one sheet, with the bias columns between the port column and the units. It has 7 pins; the bias nets are internal.

check_xschem.py compares the flat sheets with the sources with the subcircuits expanded under that naming. 0 mismatches over 12 cells. A swapped unit-net label and a changed rhigh body were both reported.

18:51 — testbench schematics and figures

Testbench schematics. xschem/tb_*.sch has one sheet per deck in sim/tb/, nine in all. They are laid out like ../xschem/tb_d2s_*.sch. The DUT and the bias come from the schematics in xschem/. Each sheet's code block holds the deck's .lib, .param and .save lines and its .control section, unchanged. check_xschem.py now also compares every testbench with its deck: each element (nets with GND = 0, value or model and sizes) and every code line.

Result: 0 mismatches over 12 cells and 9 testbenches. A changed RL value and a changed meas line were both reported. Netlisted by xschem and run in ngspice, all nine reproduce the decks' reference values: loop 78.9 dB / 1.366 MHz / 72.4°, gain 0.49994, THD 0.1187 %, noise 230 µV, units 5.442 / 5.416 µA, MOS C–V 0.370 / 0.648 / 0.745 pF, and d2s_lc2 74.0 dB / 1.49 MHz / 104.3°.

Sheets re-saved between 18:00 and 18:16. Seven sheets on the linked computer had been saved again with xschem 3.4.8RC: d2s_bias_lp, d2s_lc2, d2s_lc2_nc, d2s_mpdda, d2s_mpdda_flat, d2s_mpdda_bias_flat and unit_r2. The differences are xschem's own rewriting: merged wire segments, reversed wire directions and reordered texts. Wire geometry and instances are unchanged, and the checker passes on these versions. They were left as they are; the figures were made from them.

Figures. figures/<sheet>.svg, .pdf and .png exist for all 21 sheets (63 files). Each is xschem's own SVG export with a white background and black wires, symbols and text. Block frames and legends are #d55e00, about 47 % grey in monochrome. Each figure is cropped to the drawing and leaves out the title block, pin letters, pin squares, m=1, ng=1, b=0, the rhigh R= expression and the MOS model names. The PNGs are 2 px per unit, at most 6000 px wide; the PDFs embed Liberation Sans. Details are in xschem/README.md.

The SVG and PNG files reached the linked computer with a C2PA provenance block added in transfer: <metadata> in the SVG and a caBX chunk in the PNG. The pictures themselves are unchanged (pixel difference 0), and the PDFs arrived unchanged.

18:59 — regeneration scripts in xschem/scripts/

- export_figures.py makes the figures; by default it does all sheets of xschem/ into figures/. It needs xschem, $PDK_ROOT, cairosvg and Pillow.
- gen_cells.py OUTDIR and gen_testbenches.py OUTDIR drew the cell and testbench sheets; they share the helper xsheet.py. They refuse to write into xschem/ unless --overwrite is given, because the sheets there are now edited by hand.

Before shipping, all three ran in the cloud from a copy of the folder layout. The figures came out identical to those in figures/: the same SVG, and PNG pixels with no difference. The generators' 34 output files were byte-identical to the first drafts. The guard against writing into xschem/ also worked. The scripts have not been run on the linked computer; its sandbox has no xschem. xschem/README.md has a "Regenerating" section with the commands.

19:43 — fix: interactive plot in `tb_mpdda_noise`

Run from xschem in the IIC-OSIC-TOOLS container (ngspice-47), the noise testbench stopped with `loglog: no such command available in ngspice`. The `if $?batchmode = 0` block of the deck held `loglog onoise_spectrum`, but `loglog` is an option of `plot`, not a command. My earlier runs used `ngspice -b`, which skips that block, so the line had never been executed. The simulation results were not affected: 230.437 µV total, 676 nV/$\sqrt{\text{Hz}}$ at 1 kHz, 152 nV/$\sqrt{\text{Hz}}$ at 100 kHz.

The line now reads `plot onoise_spectrum loglog`, in `sim/tb/tb_mpdda_noise.spice` and in `xschem/tb_mpdda_noise.sch`. The sheet was edited in place: the user's 3.4.8RC save of 19:11 (xschem's normalization only) is kept. The `tb_mpdda_noise` figure was regenerated, because its code block shows the line.

Checked: all nine testbenches were run in ngspice-42 in a pseudo-terminal on a virtual display, so that the `plot` commands actually execute. The seven decks with a plot block plot without an error message, the noise spectrum on log–log axes. The unfixed deck reproduces the error. `check_xschem.py`: 0 mismatches for `tb_mpdda_noise`.

2026-10-03 16:05 — size mockup of `d2s_mpdda` (Miller compensated) in `size_mockup/`

`size_mockup/d2s_mpdda_size_mockup.py` (and its `.gds`) is made like the existing `../size_mockup/d2s_size_mockup.py`. It draws `d2s_mpdda` with its four `unit_r2` and `d2s_bias_lp`, at the `.subckt` defaults. Every device is a real SG13_dev PCell with the netlist's w/l/ng. OP and ON are the IO cell's clamp frames, as before. CMA and CMB are hv PMOS 16 × 16 µm, about 0.8 pF each at the tt operating point (V_GW ≈ 0.95 V; 0.48 pF at 0.15 V, 0.97 pF at 1.25 V, simulated at 16 × 16). The R_* are rhigh 0.5 × 50.6 µm, 150 kΩ (simulated). The grid follows `xschem/d2s_mpdda_bias_flat.sch`, with the bias on the same sheet at the left. The IO cell is placed for scale, with the same outlines as before.

Checked: the old script, run in the cloud (pip klayout 0.30.12, IHP-Open-PDK dev), reproduces the old GDS exactly (XOR empty on every layer). The new GDS holds the netlist's 42 devices outside the frames, with matching model, w, l and ng and nothing extra. A deliberately wrong size in a copy was reported.

Device bounding-box areas (PCell extents, as the old script prints them), µm²:

block	`d2s_miller`	`d2s_mpdda`
DDA: pairs, tails, rhigh	529	969
fold sinks SX, SY	152	529
fold cascodes CX, CY	49	244
mirror PL, PR	88	276
mirror cascodes PCL, PCR	66	270
class-AB ABP, ABN, FPL, FNL	64	64
front end, without frames and Miller capacitors	948	2351
Miller capacitors CMA + CMB	1980	618
front end + Miller capacitors	2928	2969
bias (`d2s_bias` / `d2s_bias_lp`)	256	212
total outside the frames	3184	3181

By this measure the revision does not save area. The MOS capacitors save 1362 µm², and the longer devices take it back: the sinks and the mirror for offset (14.4 → 5.3 mV σ), the L = 3 µm cascodes for loop gain, and the four-unit DDA. The proposal's 2387 → 2014 µm² counts drawn W·L only. Per device, the bounding box adds source/drain, contacts and the well and thick-oxide enclosures, and the new version has more devices and more fingers. Neither number includes wiring, guard rings or well spacing. By area alone, both versions fit into the 80 × 56 µm substrate-tap region above the P frame (4458 µm²).

Picture, both mockups at the same scale: `pix/2026-10-03_opus_size_mockup_old_vs_new.png` (saved 16:13; `pix/README.md` lists it).

2026-10-05 21:09 — real bias circuits (bias.md, sim/d2s_bias_ref.spice, sim/run_bias.py)

Asked for: a real bias for the driver, with (1) a reference current from an external PMOS source, (2) a reference current into an external NMOS sink, (3) a self-contained reference without resistors or BJTs, and (4) a self-contained bandgap with BJTs. At 20:44 also: take PSRR into account, including which supplies connect to which part of the circuit.

All four keep `d2s_bias_lp`'s six diodes and its outputs, so `d2s_mpdda` is unchanged. The design and all tables are in `bias.md`; the numbers come from `sim/run_bias.py` (`sim/results_bias.txt`).

- `d2s_bias_in` / `d2s_bias_out`: an input diode plus a 6-device mirror tree. Within about 1 % over all corners; 0.8–1 %/V line sensitivity; σ about 1 %.
- `d2s_bias_oa`: an Oguey–Aebischer core re-derived for sg13_hv. M10/M11 are in weak inversion (vres = U_T ln 16), M10 is cascoded, the PMOS mirror is cascoded, and the start-up senses vbr. Process $-6\ldots+7$ %, but -35 / $+33$ % over $-40\ldots125$ °C ($I \propto \mu T^2$), σ 5.1 %, 1072 µm². The existing `OgueyAebischerBias` macro (an unsized sky130 port) is not reused.
- `d2s_bias_bg`: a Banba-type current-mode bandgap with pnpMPA and rhigh, no op-amp. ±1 % over temperature and ±0.3 % over MOS corners, but -21 / $+29$ % with the rhigh corners. R1A is 10 % longer than R1B, because with equal R1 the core rested with both PNPs off at -40 °C.
- Both self-contained cores start from 0 V at all corners and settle to the DC operating point.

Device data along the way: rhigh -0.22 %/K, rppd $+0.017$ %/K, pnpMPA V_BE -1.77 mV/K at 1 µA. Pelgrom pair coefficients from the mismatch models are 10 mV·µm (hv NMOS) and 6.6 mV·µm (hv PMOS). Specific current is about 220 nA/□ for hv NMOS ($n \approx 1.35$) and about 100 nA/□ for hv PMOS ($n \approx 1.54$).

Supplies. The front end and bias are on vdd/vss; OP/ON are on separate rails vddo/vsso. The class-AB replicas RP1/RN1 have to sit on the output-stage rails. With them on the bias rails, a 50 mV offset between the rails moves I_Q by -27 / $+41$ %; on the output rails, by ±0.5 %. RN1 and ON need local substrate tap rings on vsso.

PSRR. The bias variants change the driver's supply rejection by a few dB only. The exception is the bandgap from vss at 1 MHz: -45 dB instead of -60 dB. The output-stage rails are the limit: -31 dB at 100 kHz and -12 dB at 1 MHz, with any bias. The Miller capacitors tie the output gates to vout, so only the loop gain rejects ripple on OP's or ON's source. Options: a quiet output-stage rail, or cascode (Ahuja) compensation.

The scratch scripts used during design (in the cloud) are not saved; `run_bias.py` reproduces every number.

2026-10-05 22:25 — schematics of the bias variants (xschem/d2s_bias_*, figures/d2s_bias_*)

Asked for: xschem schematics and figures of the four bias variants.

- `xschem/d2s_bias_in`, `_out`, `_oa`, `_bg` (`.sch` and `.sym`) are drawn flat from `sim/d2s_bias_ref.spice`: reference, mirror tree and the six diodes, all at transistor level. Each current branch is one column between vdd and vss. vddo and vsso come in from the right and reach only RP1 and RN1. `xschem/README.md` describes the layout.
- `xschem/scripts/gen_bias.py` draws them. Its geometric lint finds 0 faults; the sheets have 7 / 11 / 21 / 17 wire crossings without a junction.
- `xschem/check_xschem.py` now also covers the bias sheets, compared with the flattened subcircuits, and for pnpMPA it compares a, p and m. Result: 0 mismatches over all 16 cells and 9 testbenches. Two planted faults were caught: RP1's source moved to vdd, and Q2 at m=1.
- Figures: `figures/d2s_bias_*.{svg,pdf,png}`, made with `export_figures.py`, so `figures/` now holds 75 files.
- The pnpMPA symbol's netlist format calls the Tcl procedure `ev7`. xschem 3.4.4 (the cloud's Ubuntu package) does not have it, so the checks here ran with a one-line stand-in for it in a scratch `xschemrc`. That stand-in is not shipped; current xschem defines `ev7`.

2026-10-06 02:05 — bias sheets, review round (xschem/d2s_bias_*.sch)

- Christoph's hand edit of d2s_bias_bg.sch (00:30) checked: xschem's netlist is identical to that of the generated sheet, and check_xschem.py finds 0 mismatches.
- Start-up, asked how it works and whether it draws current once the core runs. It does: MS1 → MS2 carries 0.35–0.55 µA at all times (458 nA at tt, 3.3 V, 27 °C). MS2 holds ks at 0.2–1.6 mV, so MS3 is off; its drain current is 1–10 pA, mostly junction leakage (channel 4 fA at 27 °C, 3 pA at 125 °C). bias.md's "leaks 4 fA" is the channel current at 27 °C only; to be corrected. A weaker MS1 is to be revisited later.
- Tree columns reordered on request to [R] | vabp BPC BP | vabn BNC BN, first in d2s_bias_bg, then in _in, _out and _oa by copying the _bg tree (Christoph had set the frames 20 units apart).
  - iref enters _in and _out from the left at y = −560, midway between vabp and vabn.
  - In _out the iref line carries the gates of PABN, PBNC, PBN, and the vbn line those of NABP, NBPC, NBP.
  - _oa's [R] column is now _bg's (NI flipped), with _oa's PO/POC sizes.
- A preliminary edit of _in had shorted iref to vbpc at (380, −260): BPC's drain line and the iref bus ended on the same point. Christoph fixed it before the reordering.
- check_xschem.py: 0 mismatches over all sheets; no dangling wire ends or overlapping wires.
- The reordering was done with a cloud-only scratch script, not shipped.
- Not yet updated: the sheet legends ("separate rails from the right"), xschem/README.md, and figures/d2s_bias_*.

2026-10-06 05:05 — separate output-stage rails vddo / vsso in every driver cell

Asked for: vddo and vsso in d2s_lc2, d2s_lc2_nc, d2s_mpdda, d2s_mpdda_flat, d2s_mpdda_biased and d2s_mpdda_bias_flat, with symbols, testbenches and scripts to match. Port order vdd vss vddo vsso .... ONC's bulk goes to ON's tap ring. Which slot pin vsso connects to (vssio or vss_3v3) stays open. gen_cells.py, gen_testbenches.py and the IOPadDiff2Single macro schematic stay as they are.

- Christoph had already moved OP and ON (in d2s_lc2 also DP, DN and ONC's bulk) onto vddo / vsso in d2s_mpdda, d2s_lc2 and d2s_lc2_nc, and RP1 / RN1 in d2s_bias_lp, the cell missing from the list. A short between b and vabn in his first edit of d2s_lc2_nc was found by the check and fixed by him.
- Sources: d2s_mpdda.spice, both cells of d2s_lc2.spice and d2s_bias_lp.spice now have the ports and rails of the sheets.
- Sheets:
  - The pins of the four edited sheets are in port order.
  - d2s_mpdda_flat and d2s_mpdda_bias_flat are rewired: the vddo rail runs 40 above vdd, the vsso rail 20 below vss. RP2's well and RN2's bulk are rerouted so they stay on vdd and vss.
  - d2s_mpdda_biased has the two new pins.
- Symbols: seven have vddo (top) and vsso (bottom). The driver symbols' box is 40 units wider, so vout sits 40 further right.
- Testbenches, seven decks and sheets:
  - Each has its own sources, Vddo = 3.3 V and Vsso = 0 V.
  - tb_mpdda_dc's Idd is now vdd + vddo, and tb_mpdda_op also prints i(vddo).
  - All seven decks give the same results as before in ngspice. The only difference is the time of the step's maximum on its flat top: 72 ns later, at the same value.
- Scripts:
  - run_bias.py no longer builds a split copy of d2s_mpdda: split_driver() is gone, and the netlist is included as it is.
  - run_improvements.py ties vddo / vsso to vdd / vss for the cells that have them.

  - `results_improvements.txt` is reproduced exactly, and so are the driver, corners and rails sections of `results_bias.txt`.

- `check_xschem.py`: its two fixture subcircuits are updated. 0 mismatches over all sheets.
- Docs: `xschem/README.md` has the new pins, a section on the rails, and the bias sheets' current layout. `bias.md`: vddo comes from vdd_3v3; vsso's pin is open.
- Figures re-exported for the 14 changed sheets. The bias sheets and their figures are unchanged; their legends still say "from the right".
- Left in Christoph's sheets: a short dangling wire at (2160, −450…−440) in `d2s_lc2` and `d2s_lc2_nc`. It connects to nothing. Christoph removed both at 06:02. Afterwards `check_xschem.py` still finds 0 mismatches, and no sheet has a dangling wire end; figures of both re-exported.

2026-10-19 17:56 — CACE verification setup: cloud tools, checks, status

Asked for on 2026-10-06 (`../build_verification_suite/kickoff_build_verification_suite.md`): CACE testbenches for the four bias blocks alone and for each bias block with `d2s_mpdda`. At 02:20 the work was held until the scope is settled. Only the cloud tools were set up. That chat's log: `sudelbuecher/chatlog/2026-10-06_opus_class_ab_pad_driver_cace_verification_setup.md`.

- Cloud environment, 2026-10-06 02:22–03:10:
  - ngspice-42 (Ubuntu) and the IHP-Open-PDK `dev` sparse checkout at `8e07a1f` (2026-10-05).
  - psp103, psp103_nqs, r3_cmc, mosvar, cap_cmomi and cap_cmomf are compiled with OpenVAF-reloaded (`fdf2522`, LLVM 18), with the OSDI 0.4 → 0.3 patch for ngspice 42. The PDK's prebuilt `cap_cmom*.osdi` are OSDI 0.4, so they were recompiled too.
  - xschem: Ubuntu's 3.4.4 ignores `top_is_subckt`, which CACE sets. The DUT netlist then has no `.subckt` line, and every run fails with "unknown subckt". xschem master (`c15c6c0`, 3.4.8RC) is built instead. It has `ev7`, so no stand-in is needed.
  - CACE 2.13.0 under Python 3.12, because the container's Python 3.13 has no tkinter. The 2026-10-02 runs used 2.11.0.
  - This is not the IIC-OSIC-TOOLS 2026.09 pin of the improvements chat. To be aligned before real runs.
- Checks, on cloud copies of 2026-10-06 02:30 (before the 05:05 rail change):
  - `xschem/check_xschem.py`, unmodified: 0 mismatches.
  - `../run_d2s.py` reproduces `../results.txt`; only the THD comment line differs.
  - `sim/run_bias.py`, all sections, 81 s: reproduces `sim/results_bias.txt`, except the line with the OSDI path.
  - CACE, `d2s_miller_biased.yaml -s schematic -p dc_params loop_params`: about 136 of 155 runs without error, then a session restart killed it. No result.
- Status on 2026-10-19: no CACE result yet for the bias blocks or for `d2s_mpdda`. The cloud copies predate the 05:05 vddo / vsso change and have to be staged again.
- Readable CACE output in the three connected folders goes to two places:
  - `_docs/` next to the `.yaml`: CACE's own datasheet `<cell>.md` and plots. Used by `macros/OgueyAebischerBias` and `../verification/cace` (`d2s_miller`; there also the hand-written `first_run_summary.md` and `performance_figures.md`).
  - `results/<cell>/`: CSV and PNG from `scripts/cace_to_csv.py`, summarized in the README. Used by `levelshifters/ls_pair` (the newest setup, 2026-10-04, with `cace.mk`) and by `macros/inverter` (PNG only).
- Open: which output convention the new setups follow; which slot pin vsso goes to; which bias variants stay; spec limits or `any`; the harness `ibias`. Power-off is not in CACE. The power-down chat simulated it with ngspice decks.